Hugo LAM

+33 6 19 32 67 30 hugocd.lam@gmail.com www.linkedin.com/in/hugolam

French nationality - Paris, France

Objective: Seeking a 5 to 6 month internship in Mechanical Engineering starting March 2027 as part of my Master of Science in Aerodynamics and Aeroacoustics

Target Areas: External & Internal Aerodynamics, CFD, Aeroacoustics, Flow Optimization

Education

Master of Science (M2) in Engineering - Aerodynamics
Sorbonne Université - École Nationale Supérieure des Arts et Métiers (ENSAM), Paris
2026 - 2027

Master of Science (M1) in Engineering – Computational Mechanics
Sorbonne Université, Paris
2025 - 2026

Bachelor's degree – Major in Mechanics, Minor in AI & Data Science
Sorbonne Université, Paris
2022 - 2025 | **Honors : Assez Bien**

Baccalauréat Général
Lycée Pierre-Gilles de Gennes (ENCPB), Paris
2019 - 2022 | **Honors : Bien**
Majors: Mathematics and Engineering Sciences

Work Experience

Research intern
Jean Le Rond d'Alembert Institute – Paris | May 2026 – July 2026

- Numerical modeling of Rayleigh-Bénard bioconvection instabilities driven by gravity and gyrotaxis coupling, extended to phototaxis under light illumination; analyzed plume dynamics (spacing, velocities, vorticity) and geometric confinement effects
- Supervised by José-Maria Fullana and Arnaud Antkowiak (based on Anaëlle Givaudan's doctoral research)

Aircraft Maintenance Intern
Cathay Pacific – Paris-Charles de Gaulle Airport | July 2024

- Conducted arrival and transit walkaround inspections on Boeing 777-300ER and Boeing 747-400 aircraft, strictly adhering to manufacturer AMM (Aircraft Maintenance Manual) procedures
- Supported line maintenance operations: engine oil replenishment, exterior lighting checks, and cabin safety inspections

Multipurpose Team Member (Part-time — Weekends)
Taiyaki Paris – Paris | July 2024 (1 month), July 2025 (1 month)

- Handled customer service, counter sales, order processing, and store organization

Academic projects

CFD Study of a NACA 0012 Airfoil: Aerodynamic flow simulation around a NACA 0012 airfoil across subsonic and transonic Mach numbers; post-processing of aerodynamic coefficients (C_L, C_D) and benchmark comparison against wind tunnel data

Machine Learning Digit Classifier: Built and trained machine learning models in Python to maximize handwritten digit recognition accuracy; authored a comprehensive technical project report

Skills & Languages

- **Languages:** French (Native), English (B2)
- **Programming Languages:** Python, C
- **Data Analysis, Simulation & Visualization:** ParaView, Jupyter Notebook, Spyder
- **Core Competencies in Mechanics:** Continuum mechanics, Mathematical & numerical methods for mechanics, Incompressible aerodynamics, Compressible flows, Environmental fluid mechanics, Aero-elasticity